SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA1409 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: **Total Marks:50**

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I E.Ragul / 192571-32 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

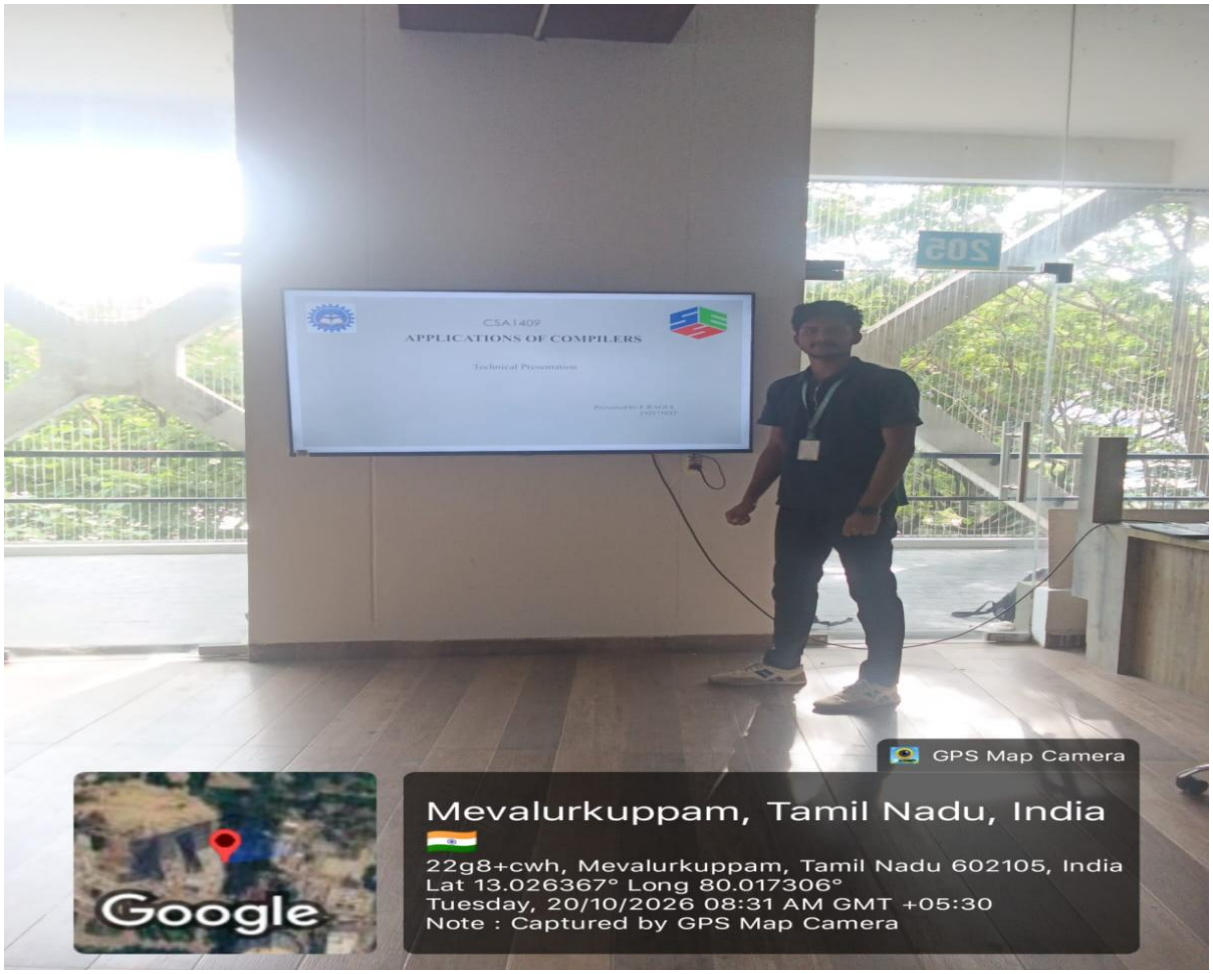
Signature of the student:
E.Ragul

Annexure A

TECHNICAL PRESENTATION

1 . Applications of Compilers

a) Software development.
b) Embedded systems.
c) Game engines.
d) AI programming.
e) System software.



CSA1409

APPLICATIONS OF COMPILERS

Technical Presentation

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1. Introduction to Compilers

- A compiler translates source code into a target language, often machine code.
- It checks the program for syntax and semantic errors.
- It may optimize code to improve execution efficiency.
- Compilation commonly includes analysis, optimization, and code generation.

Source Code → Compiler → Target Code

For example, programmers can write:

```
int a = 10;
int b = 20;
int sum = a + b;
```

2. Syntax and semantic errors

```
int a = 10
```

Source Code:

Program written in C or C++

Compiler:

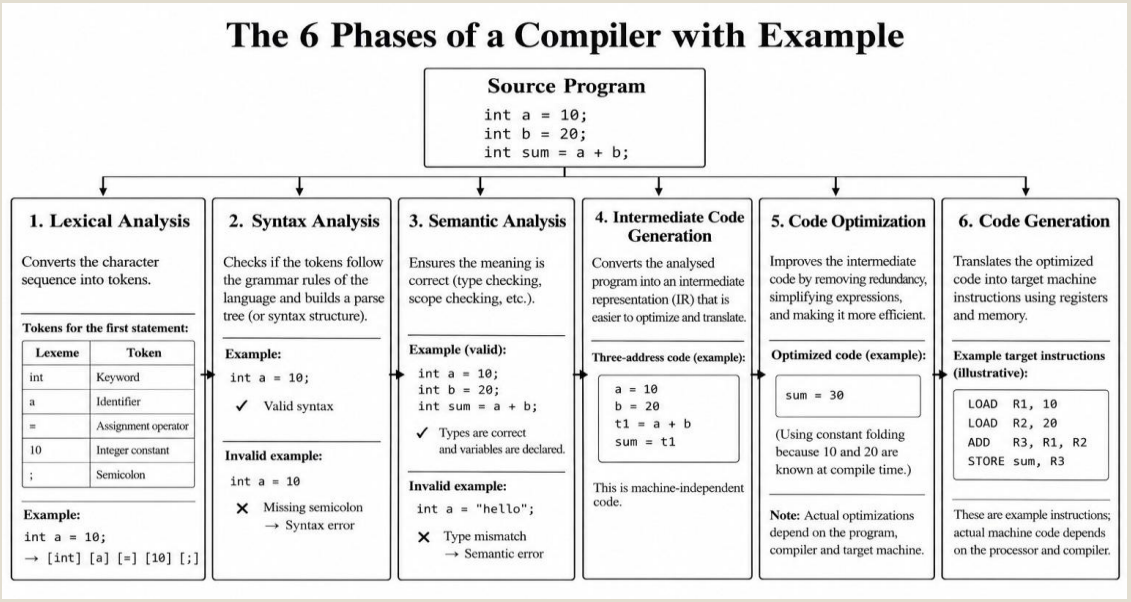
Analysis, checking and translation

Target Code:

Machine code or another target representation

2. Main Phases of a Compiler

- Lexical analysis: converts characters into tokens.
- Syntax analysis: checks grammatical structure and builds a parse tree.
- Semantic analysis: checks types, declarations, and scope.
- Intermediate code generation: creates a machine -independent representation.
- Code optimization: removes unnecessary operations where possible.
- Code generation: produces instructions for the target machine.



3. Applications in Software and Embedded Systems

- Software development: compiles programs for desktop and business applications.
- System software: supports operating -system components, drivers, and utilities.
- Embedded systems: compiles firmware for microcontrollers and dedicated devices.
- IoT devices: supports programs that control sensors and connected hardware.
- Example: C/C++ source code can be compiled for an Arduino -compatible target.

4. Applications in Games and Artificial Intelligence

- Game development: native compilation can help execute game code efficiently.
- Performance optimization: compilers can simplify operations and improve generated code.
- Artificial intelligence: compiler tools can optimize numerical and tensor operations.
- Hardware targets: code may be prepared for CPUs, GPUs, or accelerators.
- Just -in-time compilation: some systems compile code during program execution.

5. Example, Benefits and Conclusion

Example C Program

```
#include <stdio.h>
int main(void) {
   int a = 4, b = 6;
   printf("%d", a + b);
   return 0;
}
```

Key Points

- The program prints 10 after compilation and execution.
- Compilers support efficient software development and target-specific code.
- They are used in applications, embedded devices, games, and AI.
- Correctness and performance also depend on program design and testing.

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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand